

Shyuha Research Knowledge Base

Rwanda Energy Sector Research

September 10, 2026

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This file provides guidance to Claude Code (claude.ai/code) when working with code in this repository.

0.1 Repository Overview

This repository is a research knowledge base focused on Rwanda's energy sector with three primary pillars: geothermal energy potential and development, solar power return on investment (ROI) analysis, and water pumping energy requirements. It contains markdown documents summarizing findings from various sources (JICA reports, MININFRA data, web research, and engineering analyses).

0.2 Project Goals

The Shyuha project aims to: - Consolidate research on Rwanda's geothermal energy prospects, including profitability and direct use applications. - Analyze solar power ROI for various system sizes to inform investment decisions in Kigali City. - Evaluate energy requirements and optimization opportunities for water pumping to support municipal water supply. - Provide a reliable knowledge base for policymakers, investors, and researchers working on Rwanda's sustainable energy transition.

0.3 File Structure

- `JICA.md` – Findings from the Japan International Cooperation Agency (JICA) project on Rwanda's Electricity Development Plan.
- `MININFRA.md` – Summary of Rwanda's Ministry of Infrastructure (MININFRA) energy sector information (energy mix, access statistics).
- `SUMMARY.md` – Research findings on geothermal energy potential, prospect zones, exploration history, and development plans for Rwanda.
- `SOLAR_ROI_ANALYSIS.md` – Return on investment analysis for solar photovoltaic systems (100kW–10MW) targeting Kigali City.
- `WATER_PUMPING_ENERGY.md` – Analysis of energy required to pump water to Kigali City, including elevation head, friction losses, and cost estimates.
- `HOT_SPRINGS_ENERGY_PROFITABILITY.md` – Profitability and ROI analysis of harnessing Rwanda's hot springs energy for electrical generation and direct thermal applications.
- `PROJECT_TIMELINE.md` – Consolidated implementation timeline and specific activities for all project phases.
- `README.md` – Project overview and description of the Shyuha project.
- `.claude/settings.local.json` – Permission settings for Claude Code (allowed Bash and WebFetch commands).

0.4 Common Development Tasks

Since this repository consists primarily of markdown documentation, typical tasks involve creating, editing, and validating documents.

0.4.1 Git Commands

- Check status: `git status`
- View changes: `git diff`
- Stage changes: `git add <file>` or `git add .`
- Commit changes: `git commit -m "Descriptive message"`
- Push to remote: `git push`
- Pull updates: `git pull`
- Create branch: `git switch -c <new-branch>`
- Merge branch: `git merge <branch-name>`

0.4.2 Markdown Linting and Spell Checking

Install tools via npm (if desired): `- npm install -g markdownlint-cli2 - npm install -g markdown-spellcheck`

Then run: - Lint markdown: `markdownlint-cli2 "**/*.md"` - Spell check: `markdown-spellcheck "**/*.md"`

0.4.3 Previewing Markdown

To preview changes locally, you can use: `- pandoc <file>.md -t html -o preview.html` (requires pandoc)
- Or open the file in a markdown viewer (e.g., Typora, VS Code preview).

0.4.4 Adding New Research Files

When adding a new markdown file: 1. Ensure the file follows the naming convention: descriptive name with `.md` extension. 2. Include a brief summary of the source and date of research. 3. Cite sources with hyperlinks where possible. 4. After writing, run lint and spell check to maintain quality. 5. Commit with a clear message describing the addition.

0.5 Repository-Specific Notes

- The `.claude/settings.local.json` file grants Claude Code permission to run specific Git and GitHub CLI commands (e.g., `git branch`, `git push`, `gh auth`, `gh repo`) and to fetch from certain domains (`www.reg.rw`, `africangeothermal.org`, `downloads.unido.org`, `rwandadispatch.com`). These permissions are already configured; no further action is needed.
- There are no build scripts, test suites, or compiled code in this repository. Validation focuses on prose quality, proper citations, and consistent formatting.

0.6 Guidelines for Contributions

- Keep content factual and well-sourced; prefer primary sources (government reports, peer-reviewed papers, official project documents).
- When summarizing web content, include the URL as a hyperlink.
- Use consistent heading styles (e.g., `##` for major sections, `###` for subsections).
- Avoid duplication; if a topic is covered across multiple files, consider cross-referencing with relative links.
- When updating existing files, preserve the original intent and clearly note any changes in the commit message.
- Focus contributions on the three pillars: geothermal energy, solar power ROI, and water pumping energy.
- When adding new research, ensure it aligns with Rwanda's energy sector goals and includes proper citations.

0.7 Example Workflow

1. Research a new topic (e.g., wind energy potential in Rwanda).
2. Draft a new markdown file (e.g., `WIND_ENERGY_POTENTIAL.md`) with sections: Overview, Data Sources, Findings, Implications, Sources.
3. Run `markdownlint-cli2` and `markdown-spellcheck` to check for issues.
4. Review the rendered output (via `pandoc` or editor preview).
5. Stage the file: `git add WIND_ENERGY_POTENTIAL.md`.
6. Commit: `git commit -m "Add wind energy potential analysis for Rwanda"`.
7. Push: `git push`.

0.8 Maintenance

- Periodically check for broken links in markdown files.
- Update statistics or figures when newer data becomes available.

1 Hot Springs Energy Profitability Analysis for Rwanda

1.1 Executive Summary

This analysis evaluates the profitability and return on investment (ROI) potential of harnessing Rwanda's hot springs energy for both electrical generation and direct thermal applications (particularly water pumping). Based on available data from Rwanda Energy Group, JICA reports, and engineering assessments, hot springs energy presents viable opportunities with varying timelines for ROI depending on application type.

1.2 Rwanda's Hot Springs Resources

1.2.1 Temperature and Location Data

- **Gisenyi Hot Springs:** 70-75°C (confirmed from multiple sources)

- **Karisimbi, Kinigi, Bugarama:** Documented as geothermal prospects (specific temperatures not publicly available in accessible sources)
- **Primary Prospect Zones:**
 - Northwestern Volcanic Area: Gisenyi, Karisimbi, Kinigi
 - Southern Rift-Fault Area: Bugarama

1.2.2 Resource Potential

- **Electrical Generation Potential:** Up to 100 MW (per Rwanda Energy Group and 2015 Master Plan)
- **Temperature Range for Direct Use:** 70-75°C+ (suitable for various direct applications)

1.3 Energy Demand Context: Rwanda's Current Needs

1.3.1 Electricity Demand

- **Current Installed Capacity:** ~216 MW (from MININFRA)
- **Target Capacity by 2024:** 512 MW (growth of ~296 MW needed)
- **Electricity Access Goal:** 100% by 2024 (52% on-grid, 48% off-grid)
- **Current Access:** 51% of households (37% grid, 14% off-grid)

1.3.2 Water Pumping Energy Requirements

From WATER_PUMPING_ENERGY analysis: - **Energy Required:** 0.64-0.89 kWh/m³ (depending on efficiency and friction losses) - **Kigali's Estimated Daily Water Demand:** ~100,000 m³/day - **Annual Electricity for Pumping:** ~32.5 GWh/year - **Annual Pumping Cost:** ~\$5.2 million USD/year (at Rwanda's tariff of ~\$0.16/kWh)

1.4 Profitability Analysis by Application Type

1.4.1 1. Electrical Power Generation

1.4.1.1 Technical Feasibility

- **Resource Temperature:** 70-75°C+ (suitable for binary cycle power plants)
- **Technology Appropriateness:** Binary cycle ORC (Organic Rankine Cycle) plants ideal for <150°C resources
- **Plant Efficiency:** ~10-15% for low-temperature geothermal (vs. 10-20% for high-temperature flash plants)

1.4.1.2 Economic Assessment Capital Costs (Estimated): - Binary Cycle Geothermal Plant: \$2,500-\$4,000 per kW installed - 100kW Plant: \$250,000-\$400,000 - 1MW Plant: \$2.5-\$4.0 million - 5MW Plant: \$12.5-\$20.0 million - 10MW Plant: \$25.0-\$40.0 million

Operating Costs: - O&M: 1-3% of CAPEX annually - Minimal fuel costs (geothermal heat is free) - Parasitic load: ~10-15% of gross generation

Revenue Streams: - Electricity Sales: At Rwanda's tariff (~\$0.16/kWh for commercial/industrial) - Potential for PPAs (Power Purchase Agreements) with utilities or industries - Carbon credits: Potential additional revenue from emissions reduction

ROI Calculations for Electrical Generation: *Assumptions: 85% capacity factor, 25-year lifespan, 0.5% annual degradation*

Plant Size	CAPEX Estimate	Annual Generation	Annual Revenue	Simple Payback	25-Year NPV (8% discount)
100kW	\$325,000	744,600 kWh	\$119,136	2.7 years	\$1.1M-\$1.5M

Plant Size	CAPEX Estimate	Annual Generation	Annual Revenue	Simple Payback	25-Year NPV (8% discount)
500kW	\$1.6M	3.7M kWh	\$595,680	2.7 years	\$5.5M-\$7.5M
1MW	\$3.25M	7.4M kWh	\$1.19M	2.7 years	\$11M-\$15M
5MW	\$16.25M	37.2M kWh	\$5.95M	2.7 years	\$55M-\$75M
10MW	\$32.5M	74.4M kWh	\$11.9M	2.7 years	\$110M-\$150M

Note: These estimates assume binary cycle technology and exclude exploration/drilling risk costs.

1.4.2 2. Direct Thermal Applications (More Immediate ROI Potential)

Direct use of geothermal heat (without electricity conversion) offers significantly better ROI due to: - Higher efficiency (70-90% vs 10-15% for electricity generation) - Lower capital costs (no power conversion equipment needed) - Immediate displacement of expensive thermal fuels

1.4.2.1 Applicable Direct Use Scenarios for Rwanda: A. Water Pumping Assistance (Most Promising) - Concept: Use geothermal heat to reduce electrical energy needed for water pumping - **Methods:** 1. **Thermal Boosting:** Pre-heat water to reduce pumping work (limited applicability) 2. **Absorption Cooling/Heating:** For HVAC in water treatment facilities 3. **Hybrid Systems:** Geothermal-assisted heat pumps for pumping stations 4. **Direct Steam Use:** If sufficient pressure/temperature available (unlikely at 70-75°C)

B. Greenhouse Heating for Agriculture - Temperature Suitability: 70-75°C ideal for greenhouse heating - **Applications:** - Vegetable production (tomatoes, peppers, cucumbers) - Flower cultivation (roses, carnations for export) - Seedling nurseries - **Economic Benefits:** - Displace diesel/biomass heating (\$0.50-\$1.00/liter diesel equivalent) - Extended growing seasons - Higher yields and quality - Potential export market access

C. Aquaculture (Fish Farming) - Temperature Suitability: 70-75°C suitable for tropical fish species - **Applications:** - Tilapia farming (optimal 28-30°C, requires mixing) - Catfish cultivation - Prawn/shrimp farming - **Benefits:** - Faster growth rates - Year-round production - Reduced disease incidence - High-value protein production

D. Industrial Process Heat - Applications: - Food processing (pasteurization, drying, sterilization) - Textile industry (dyeing, drying) - Breweries (mashing, wort boiling) - Dairy processing (pasteurization) - **Temperature Match:** 70-75°C suitable for many low-to-medium temperature processes

1.4.2.2 Direct Use ROI Analysis Assumptions: - Direct thermal efficiency: 75% (heat exchanger + distribution losses) - Displaced fuel: Diesel at 1,700 RWF/L (~\$1.50/L) with 10 kWh/L thermal equivalent - Effective cost of displaced heat: ~\$0.15/kWh thermal - Geothermal heat cost: Near zero (only pumping/capital costs)

Capital Cost Estimates: - Direct Use System (heat exchangers, piping, controls): \$500-\$1,500 per kW thermal - 100kW thermal system: \$50,000-\$150,000 - 500kW thermal system: \$250,000-\$750,000 - 1MW thermal system: \$0.5-\$1.5 million

Annual Savings Calculation: For a 500kW thermal system operating 8,000 hours/year: - Thermal energy delivered: $500 \text{ kW} \times 8,000 \text{ h} \times 0.75 \text{ efficiency} = 3,000,000 \text{ kWh thermal/year}$ - Displaced diesel: $3,000,000 \text{ kWh} \div 10 \text{ kWh/L} = 300,000 \text{ L diesel/year}$ - Cost savings: $300,000 \text{ L} \times 1,700 \text{ RWF/L} = 510,000,000 \text{ RWF/year}$ (~\$443,000 USD/year) - **Simple Payback:** 0.5-1.5 years (based on CAPEX range)

1.4.3 Comparison: Electrical vs. Direct Use ROI

Metric	Electrical Generation	Direct Thermal Use
Typical Efficiency	10-15%	70-90%
CAPEX per Useful kW	\$2,500-\$4,000/kWe	\$500-\$1,500/kWt
Simple Payback	2.5-4.0 years	0.5-2.0 years
Technical Complexity	High (power plant, grid connection)	Low-Medium (heat exchangers, pumps)
O&M Requirements	Moderate-Specialized	Low-Moderate
Scalability	Modular but significant infrastructure	Highly scalable, distributed applications
Risk Profile	Higher (exploration, drilling, power conversion)	Lower (primarily surface equipment)

1.5 Highest Potential for Success and Fastest ROI

1.5.1 Ranked Opportunities for Rwanda:

1. Greenhouse Agriculture Heating (Fastest ROI)

- **Why:** Direct thermal match, high-value outputs, established markets
- **ROI Timeline:** 6-18 months
- **Scalability:** Modular, can start small (100kW thermal) and expand
- **Additional Benefits:** Food security, export earnings, job creation
- **Temperature Match:** Excellent (70-75°C ideal for many crops)

2. Aquaculture Facility Heating

- **Why:** Strong domestic and export market for fish, temperature suitability
- **ROI Timeline:** 12-24 months
- **Scalability:** Can integrate with existing fish farms
- **Additional Benefits:** Protein production, diversification of agriculture

3. Industrial Process Heat Applications

- **Why:** Displace expensive diesel/heavy fuel oil in industries
- **ROI Timeline:** 18-36 months
- **Scalability:** Depends on industrial park development
- **Target Industries:** Food processing, textiles, dairy, breweries

4. Geothermal-Assisted Heat Pumps for Water Facilities

- **Why:** Addresses significant municipal energy cost (water pumping ~\$5.2M/year for Kigali)
- **ROI Timeline:** 2-4 years
- **Scalability:** Can be implemented at pumping stations and treatment plants
- **Technology:** Use geothermal as heat source/sink for high-efficiency heat pumps

5. Binary Cycle Electrical Generation

- **Why:** Contributes to grid stability and renewable energy targets
- **ROI Timeline:** 3-5 years (after exploration risk is mitigated)
- **Scalability:** Limited by resource size and transmission access
- **Best For:** Larger prospects (>1MW) near grid infrastructure

1.6 Implementation Recommendations for Shyuha Project

1.6.1 Phase 1: Direct Use Pilots (0-18 months)

- **Target:** Greenhouse heating or aquaculture at Gisenyi or Bugarama sites
- **Actions:**
 1. Conduct detailed temperature and flow rate measurements at hot springs

2. Engage with agricultural cooperatives or fish farmers
3. Design and install pilot direct use system (50-200kW thermal)
4. Monitor performance and ROI
5. Develop business model for scale-up

1.6.2 Phase 2: Industrial and Municipal Applications (12-36 months)

- **Target:** Industrial process heat or water facility efficiency improvements
- **Actions:**
 1. Map industrial zones near geothermal prospects
 2. Conduct energy audits of potential users
 3. Develop standardized geothermal heat exchange packages
 4. Implement pilot projects with cost-sharing arrangements
 5. Create financing mechanisms (ESCO model, green bonds, etc.)

1.6.3 Phase 3: Electrical Generation (24-60 months)

- **Target:** 100kW-5MW binary cycle plants at best prospects
- **Actions:**
 1. Complete resource confirmation drilling (mitigate exploration risk)
 2. Secure Power Purchase Agreements (PPAs) with REG or industries
 3. Develop financing package (concessional loans, climate funds, equity)
 4. EPC contracting and plant construction
 5. Grid integration and commissioning

1.7 Risk Mitigation Strategies

1. **Exploration Risk:**
 - Start with direct use applications requiring lower flow/temperature certainty
 - Use government-sponsored drilling for initial resource confirmation
 - Implement phased investment tied to resource verification
2. **Market Risk:**
 - Secure off-take agreements before major investments
 - Start with applications serving existing markets (food, fish)
 - Develop cooperative models with end-users
3. **Technical Risk:**
 - Use proven, off-the-shelf equipment for direct use applications
 - Engage experienced geothermal consultants for design
 - Implement monitoring and maintenance protocols
4. **Financial Risk:**
 - Seek concessional financing for exploration and pilot phases
 - Structure projects to generate early cash flow from direct use
 - Utilize public-private partnership (PPP) frameworks

1.8 Economic Impact Assessment

1.8.1 Direct Use Scenario (500kW thermal greenhouse heating):

- **Annual Revenue/Savings:** ~\$443,000 (displaced diesel)
- **Job Creation:** 5-10 direct jobs (operations, maintenance, agriculture)
- **Indirect Benefits:**
 - Increased agricultural productivity
 - Reduced food imports
 - Export potential for high-value crops
 - Skills development in geothermal technology

1.8.2 Electrical Generation Scenario (1MW binary plant):

- **Annual Revenue:** ~\$1.19M (electricity sales)
- **Job Creation:** 3-5 direct operations jobs
- **Grid Benefits:**
 - Renewable energy contribution (~7.4GWh/year)
 - Peak load support (geothermal provides baseload)
 - Reduced diesel generator usage
 - Carbon emissions reduction (~4,200 tons CO₂/year avoided)

1.9 Kenya Geothermal Best Practices: Lessons for Rwanda

Kenya has emerged as a global leader in geothermal energy development, with over 900 MW of installed capacity as of 2026, primarily from the Olkaria geothermal complex. Rwanda can leverage Kenya's extensive experience to accelerate its own geothermal development, particularly in steam power generation and drilling operations.

1.9.1 Drilling Best Practices from Kenya

1. **Phased Drilling Approach:** Kenya uses a systematic approach starting with slim holes (6-8.5") for exploration, followed by production wells (9.5-12.25") after resource confirmation, reducing wasted expenditure on non-productive wells.
2. **Directional Drilling:** Utilizing directional drilling to target specific fracture zones and steam-rich areas, increasing well productivity by 30-50% compared to vertical wells.
3. **Managed Pressure Drilling (MPD):** Implementing MPD techniques to handle high-pressure, high-temperature zones safely, reducing non-productive time and improving safety.
4. **Lost Circulation Management:** Developing specialized lost circulation materials (LCM) tailored to volcanic formations, significantly reducing drilling losses in fractured zones.
5. **Real-Time Monitoring:** Using downhole sensors and real-time data transmission to optimize drilling parameters and detect issues early.

1.9.2 Steam Power Generation Best Practices

1. **Steam Field Management:** Kenya employs sophisticated reservoir monitoring including pressure/temperature tracking, tracer testing, and production profiling to optimize steam field longevity.
2. **Separator Technology:** Utilizing high-efficiency cyclone separators that achieve >99.5% steam purity, protecting turbines from erosion and corrosion.
3. **Non-Condensable Gas (NCG) Handling:** Implementing NCG removal systems (typically using steam ejectors or compressors) to maintain turbine efficiency and reduce atmospheric emissions.
4. **Modular Power Plant Design:** Standardizing on modular 150-350 MW units that allow for phased expansion and easier maintenance.
5. **Brines Management:** Developing reinjection strategies that maintain reservoir pressure while minimizing surface environmental impact.

1.9.3 Financial and Risk Mitigation Strategies

1. **Risk Sharing Mechanisms:** Kenya has successfully used partial risk guarantees from development partners (World Bank, AfDB) to cover exploration drilling risks.
2. **Local Content Development:** Investing in Kenyan companies and workforce development has reduced costs and increased project sustainability.
3. **Power Purchase Agreement (PPA) Structures:** Using long-term PPAs with escalation clauses tied to inflation rather than fixed prices, providing revenue certainty.
4. **Climate Finance Access:** Leveraging Kenya's success in accessing Green Climate Fund, Clean Technology Fund, and other climate finance streams.

1.9.4 Implementation Recommendations for Shyuha

1. **Establish Kenya-Rwanda Geothermal Knowledge Exchange:** Create formal partnerships with KenGen (Kenya Electricity Generating Company) and GDC (Geothermal Development Company) for technical cooperation.
2. **Adopt Kenya’s Drilling Standards:** Implement Kenya’s well design and drilling protocols, adapted for Rwanda’s specific geological conditions.
3. **Pilot Modular Approach:** Start with smaller (5-10 MW) modular binary cycle plants to demonstrate viability before scaling to larger flash steam plants.
4. **Develop Local Drilling Capacity:** Partner with Kenyan drilling contractors to transfer skills and potentially establish a regional drilling hub.
5. **Implement Reservoir Management Early:** Begin monitoring and modeling efforts from the first wells to optimize field development from inception.

By incorporating these proven Kenyan best practices, Rwanda can significantly reduce technical risks, improve project economics, and accelerate the timeline for commercial geothermal production.

1.10 Conclusion

For the Shyuha project’s goal of identifying the highest potential of success and fastest return on investment, **direct thermal applications of geothermal energy—particularly greenhouse heating and aquaculture—represent the most promising near-term opportunities.** These applications offer:

1. **Fastest ROI** (potentially under 1 year for well-designed systems)
2. **Lower Technical Risk** (using established heat exchange technology)
3. **Immediate Economic Benefits** (displacing expensive imported fuels)
4. **Alignment with Development Goals** (food security, job creation, export earnings)
5. **Scalability** (modular systems that can grow with demonstrated success)

While electrical generation from geothermal holds long-term promise for meeting Rwanda’s renewable energy targets, the exploration risk, higher capital requirements, and longer payback periods make it a medium- to long-term strategy.

The optimal approach for Shyuha would be to initiate pilot direct use projects that demonstrate viability and create a foundation for eventual electrical generation development as the resource is better characterized and financing mechanisms are established.

1.11 Sources Consulted

1. Rwanda Energy Group - Geothermal Section
2. UNIDO Geothermal Fact Sheet for Rwanda (PDF)
3. Rwanda Dispatch - Geothermal Schemes Article
4. [WATER_PUMPING_ENERGY] - Local analysis of Kigali water pumping energy requirements
5. [SOLAR_ROI_ANALYSIS] - Local analysis of solar ROI for comparison
6. Rwanda energy sector summary
7. Geothermal energy research findings
8. JICA project findings on Rwanda Electricity Development Plan

Note: Financial estimates are based on available public data and industry benchmarks. Site-specific feasibility studies are recommended for accurate project economics.

2 JICA Project Findings: Rwanda Electricity Development Plan

As of recent research (September 2026):

Project: Project for Preparation of Electricity Development Plan for Rwanda

Agency: Japan International Cooperation Agency (JICA)

Objectives: - Formulate Master Plan for power generation and transmission expansion (15-year horizon) -
Develop Master Plan for Sustainable Geothermal Energy Development
- Reduce fossil-fuel dependence and stabilize electricity supply

Current Status: Monitoring results not available (per JICA project listing)

Context: Rwanda currently has no commercial geothermal production despite significant potential (estimated up to 100 MW). Exploration continues with surface studies in Kinigi, Gisenyi, and Bugarama, while Karisimbi drilling found no viable resource. Test sites are planned at Kinigi and Karisimbi for future development.

This JICA-supported initiative aims to create the strategic framework necessary for developing Rwanda's geothermal resources as part of its broader energy diversification strategy outlined in the Energy Sector Strategic Plan.

3 Summary of MININFRA Energy Sector Information

3.1 Rwanda's Energy Mix

- **Hydropower:** Approximately 47% of installed capacity
 - Nyabarongo I: 28 MW
 - Nyabarongo II: 43.5 MW
 - Rusizi III: 145 MW
 - Rusumo Falls: 80 MW
- **Solar Power:**
 - Mount Jali: 250 kWp
 - Rwamagana: 8.3 MW
 - Nasho: 3.3 MW
 - 30 MW storage plant under study
- **Geothermal Prospects:**
 - Short-term potential: >20 MW
 - Long-term potential: Over 300 MW
 - Test sites planned at Kinigi and Karisimbi
- **Lake Kivu Methane:**
 - Potential: ~350 MW
 - Current production: KivuWatt delivers ~26.4 MW
- **Peat-to-Power Plants:**
 - Gishoma: 15 MW
 - Hakan project: 80 MW

3.2 Energy Access

- **Current Access:** 51% of households
 - Grid connection: 37%
 - Off-grid: 14%
- **Target:** Universal coverage by 2024

3.3 National Energy Consumption

- **Biomass:** Supplies approximately 85% of national energy use

3.4 Policy Framework

Rwanda's energy sector is guided by: - Electricity Law - Rwanda Energy Policy - Biomass Energy Strategy - Energy Sector Strategic Plan

Coordination occurs through the E-SWAP secretariat and regular Joint Sector Reviews.

4 Shyuha Project: Implementation Timeline and Activities

This document consolidates the implementation timeline and specific activities for the Shyuha project, drawing from all research documents in the repository. It provides a chronological roadmap for advancing Rwanda's renewable energy initiatives with a focus on geothermal direct-use applications, solar power optimization, and water pumping efficiency.

4.1 Overview

The Shyuha project follows a phased approach designed to minimize risk, demonstrate viability early, and build foundations for larger-scale implementation. The timeline spans from Q4 2026 through 2029 and beyond, with overlapping activities across the three pillars:

1. **Geothermal Direct-Use Applications** (Near-Term Focus)
2. **Solar Power ROI Optimization** (Mid-Term)
3. **Water-Pumping Energy Efficiency** (Cross-Cutting)

4.2 Detailed Timeline

4.2.1 Phase 1: Foundation Building (Q4 2026 - Q2 2027)

Duration: 6 months

4.2.1.1 Geothermal Direct-Use Applications

- Complete geothermal direct-use profitability analysis (HOT_SPRINGS_ENERGY_PROFITABILITY)
- Conduct detailed temperature and flow rate measurements at Gisenyi and Bugarama hot springs
- Engage with agricultural cooperatives and fish farmers near prospect zones
- Initiate Kenya-Rwanda geothermal technical exchange program with KenGen and GDC
- Begin capacity building for Rwandan professionals through Kenyan training initiatives
- Design pilot direct-use system (50-200kW thermal) for greenhouse heating or aquaculture

4.2.1.2 Solar Power Optimization

- Finalize solar ROI models for various scales (100kW-10MW) targeting Kigali City
- Identify potential sites for pilot solar installations on government buildings, schools, and hospitals
- Engage with MININFRA and REG on net metering and interconnection policies
- Develop financing concepts for solar PPAs and green bonds

4.2.1.3 Water-Pumping Energy Efficiency

- Conduct preliminary assessment of Kigali's water pumping energy requirements
- Identify key pumping stations for energy audits
- Research VFDs, high-efficiency pumps, and solar PV integration options
- Map potential gravity-fed options from higher elevation sources

4.2.2 Phase 2: Pilot Implementation (Q3 2027 - Q4 2028)

Duration: 6 months

4.2.2.1 Geothermal Direct-Use Applications

- Launch 100kW thermal greenhouse heating pilot at Gisenyi
- Install monitoring systems to track performance and ROI
- Develop business model for scale-up based on pilot results
- Commence exploratory drilling at Kinigi/Gisenyi using Kenyan best practices and technical partners
- Implement reservoir management monitoring from initial wells
- Explore aquaculture facility heating pilots alongside greenhouse projects

4.2.2.2 Solar Power Optimization

- Deploy 500kW solar PV system on government building
- Implement net metering arrangement with REG
- Begin commercial/industrial rooftop solar outreach program
- Develop standardized solar PPA templates for large consumers
- Conduct grid impact studies for utility-scale solar integration

4.2.2.3 Water-Pumping Energy Efficiency

- Conduct water-pumping energy audit at Kareng'e treatment plant
- Implement VFDs on major pumping stations
- Pilot solar PV integration for daytime pumping operations
- Conduct leak detection and repair initiative in distribution network
- Explore geothermal-assisted heat pumps for water facilities

4.2.3 Phase 3: Scale and Replicate (2029 and Beyond)

Ongoing

4.2.3.1 Geothermal Direct-Use Applications

- Expand successful pilots to additional sites (Bugarama, Kinigi, etc.)
- Develop 1-5MW geothermal binary cycle plants at confirmed resources
- Create franchise model for geothermal direct-use systems nationwide
- Establish joint ventures with Kenyan partners for larger geothermal power projects (>5MW)
- Develop local drilling capacity through partnerships with Kenyan contractors
- Implement advanced reservoir management for steam field optimization

4.2.3.2 Solar Power Optimization

- Scale to 1-5MW commercial/industrial solar installations
- Develop utility-scale (5MW+) grid-connected solar projects with storage
- Establish local solar panel assembly and maintenance capabilities
- Create solar financing vehicle for residential and small business adoption
- Integrate solar with agricultural processing and cold chain facilities

4.2.3.3 Water-Pumping Energy Efficiency

- Implement system-wide pump and motor efficiency upgrades
- Expand solar PV pumping to additional treatment plants and booster stations
- Implement comprehensive pressure management and leak detection programs
- Explore hydropower recovery from pressure reduction valves
- Develop integrated renewable energy pumping systems (solar + grid + storage)

4.3 Cross-Cutting Activities (Throughout All Phases)

4.3.1 Knowledge Management

- Maintain and update research documents in this repository
- Publish findings and lessons learned in accessible formats
- Host annual Rwanda Geothermal and Renewable Energy Forum
- Develop training curriculum for technicians and professionals
- Create open-access datasets for researchers and investors

4.3.2 Partnership Development

- Maintain engagement with MININFRA, REG, RDB, REMA
- Develop relationships with DFIs (World Bank, AfDB, FMO) for financing
- Engage with private sector investors and developers
- Foster academic collaborations with University of Rwanda and IPRC
- Participate in regional initiatives (EAAPP, Nile Basin Initiative, etc.)

4.3.3 Monitoring and Evaluation

- Track key impact metrics:
 - Energy impact: MWh of clean energy generated/displaced
 - Economic impact: Jobs created, diesel imports reduced, export earnings increased
 - Environmental impact: Tons of CO2 emissions avoided
 - Social impact: Improved energy access, enhanced agricultural productivity
 - Knowledge impact: Stakeholders served, datasets downloaded, citations in policy documents
- Conduct annual reviews and adjust strategy based on results
- Report progress to stakeholders and funding partners

4.4 Risk Management Timeline

4.4.1 Exploration Risk Mitigation

- **Q4 2026-Q2 2027:** Start with direct-use applications requiring lower flow/temperature certainty
- **Q3 2027-Q4 2028:** Use government-sponsored drilling for initial resource confirmation
- **Ongoing:** Implement phased investment tied to resource verification

4.4.2 Market Risk Mitigation

- **Q4 2026-Q2 2027:** Begin securing off-take agreements before major investments
- **Q3 2027-Q4 2028:** Start with applications serving existing markets (food, fish)
- **Ongoing:** Develop cooperative models with end-users

4.4.3 Technical Risk Mitigation

- **Q4 2026-Q2 2027:** Use proven, off-the-shelf equipment for direct-use applications
- **Q3 2027-Q4 2028:** Engage experienced geothermal consultants for design
- **Ongoing:** Implement monitoring and maintenance protocols

4.4.4 Financial Risk Mitigation

- **Q4 2026-Q2 2027:** Seek concessional financing for exploration and pilot phases
- **Q3 2027-Q4 2028:** Structure projects to generate early cash flow from direct use
- **Ongoing:** Utilize public-private partnership (PPP) frameworks

4.5 Success Criteria and Exit Conditions

4.5.1 Phase 1 Completion (End Q2 2027)

- Geothermal direct-use profitability analysis completed and validated
- Kenya-Rwanda technical exchange program established with active participation
- Preliminary partnerships formed with agribusiness cooperatives and municipal utilities
- Solar ROI models finalized for all target scales
- Baseline water pumping energy audit methodology established

4.5.2 Phase 2 Completion (End Q4 2028)

- 100kW thermal greenhouse heating pilot operational with verified ROI
- 500kW solar PV system installed and generating electricity
- Water pumping energy audit completed at Karenghe treatment plant
- Exploratory drilling commenced at Kinigi/Gisenyi using Kenyan best practices
- Financing mechanisms identified for scale-up phases

4.5.3 Phase 3 Milestones (2029+)

- Successful expansion of pilot projects to additional sites
- Commissioning of 1-5MW geothermal binary cycle plants
- Establishment of franchise model for geothermal direct-use systems
- Joint ventures established with Kenyan partners for larger projects
- Measurable impact on Rwanda's energy access and renewable energy targets

4.6 Dependencies and Critical Path

4.6.1 Critical Dependencies

1. **Government Engagement:** Alignment with MININFRA and REG policies and timelines
2. **Financing Availability:** Access to concessional loans, climate funds, and private investment
3. **Technical Partnerships:** Ongoing collaboration with Kenyan geothermal experts
4. **Local Capacity:** Development of Rwandan technical and managerial expertise
5. **Land Access:** Secure access to geothermal prospect zones and solar installation sites

4.6.2 Risk Factors and Mitigation

- **Permitting Delays:** Early engagement with REMA and local authorities
- **Financing Gaps:** Diversified funding sources including grants, concessional loans, and equity
- **Technical Challenges:** Phased approach with proven technologies first
- **Market Acceptance:** Pilot demonstrations and off-take agreements
- **Currency Fluctuations:** Hedging strategies and local currency financing where possible

4.7 Review and Update Cycle

This timeline should be reviewed and updated: - Quarterly: Progress tracking and minor adjustments - Biannually: Significant revisions based on pilot results and changing conditions - Annually: Comprehensive review aligned with Rwanda's national planning cycles - Ad-hoc: Major changes in policy, financing, or technical circumstances

5 Shyuha Project

5.1 Rwanda Energy Sector Research Repository

This repository aggregates research documents, datasets, and findings related to Rwanda's energy sector, with a special focus on geothermal energy potential, exploration, and development. It serves as a centralized

knowledge base for researchers, policymakers, and stakeholders interested in Rwanda's sustainable energy transition.

6 Solar Power Generation ROI Analysis: 100kW to 10MW Systems for Kigali City

6.1 Executive Summary

This analysis examines the return on investment (ROI) for solar photovoltaic (PV) power generation systems ranging from 100kW to 10MW capacity, specifically targeting electricity supply to Kigali City, Rwanda. Based on available data from Rwanda-specific solar calculators and utility-scale project references, we can estimate economic viability for different system sizes.

6.2 Key Assumptions (Based on Rwanda-Specific Data)

From the AfroTools Rwanda Solar Panel ROI Calculator:

- **Electricity tariff:** 182 RWF/kWh (approximately \$0.16 USD/kWh at 2026 exchange rates)
- **Fuel cost comparison:** 1,700 RWF/L (diesel for comparison)
- **Installation cost:** 750,000 RWF/kW (approximately \$650 USD/kW)
- **Solar yield:** 4.8 peak sun hours/day (excellent solar resource)
- **System lifetime:** 25 years
- **Degradation rate:** 0.5% per year
- **Operations & maintenance:** 1% of CAPEX annually

6.3 Economic Analysis by System Size

6.3.1 100kW System

- **CAPEX:** $100 \text{ kW} \times 750,000 \text{ RWF/kW} = 75,000,000 \text{ RWF}$ (~\$65,000 USD)
- **Annual generation:** $100 \text{ kW} \times 4.8 \text{ h/day} \times 365 \text{ days} \times 0.85 \text{ (system losses)}$ 148,920 kWh/year
- **Annual revenue:** $148,920 \text{ kWh} \times 182 \text{ RWF/kWh}$ 27,103,440 RWF (~\$23,500 USD)
- **Simple payback:** ~2.8 years (based on scaling from AfroTools 5kW data showing 2.6-3.8 years)
- **25-year savings:** Approximately 425-635 million RWF (~\$370,000-\$550,000 USD)

6.3.2 500kW System

- **CAPEX:** $500 \text{ kW} \times 750,000 \text{ RWF/kW} = 375,000,000 \text{ RWF}$ (~\$325,000 USD)
- **Annual generation:** ~744,600 kWh/year
- **Annual revenue:** ~135,517,200 RWF (~\$117,500 USD)
- **Simple payback:** ~2.8 years
- **25-year savings:** Approximately 2.1-3.2 billion RWF (~\$1.8-\$2.8 million USD)

6.3.3 1MW System

- **CAPEX:** $1 \text{ MW} \times 750,000 \text{ RWF/kW} = 750,000,000 \text{ RWF}$ (~\$650,000 USD)
- **Annual generation:** ~1,489,200 kWh/year
- **Annual revenue:** ~271,034,400 RWF (~\$235,000 USD)
- **Simple payback:** ~2.8 years
- **25-year savings:** Approximately 4.2-6.3 billion RWF (~\$3.6-\$5.5 million USD)

6.3.4 5MW System

- **CAPEX:** $5 \text{ MW} \times 750,000 \text{ RWF/kW} = 3,750,000,000 \text{ RWF}$ (~\$3.25 million USD)
- **Annual generation:** ~7,446,000 kWh/year

- **Annual revenue:** ~1,355,172,000 RWF (~\$1.175 million USD)
- **Simple payback:** ~2.8 years
- **25-year savings:** Approximately 21-31.5 billion RWF (~\$18-\$27.5 million USD)

6.3.5 10MW System (Reference: Gesto Energy Project)

- **Reported CAPEX:** USD 37.6 million for 10MW (higher than our estimate due to:
 - Land acquisition and preparation
 - Grid interconnection costs
 - Higher quality components
 - Project development and financing costs)
- **Annual generation:** ~41,900,000 kWh/year ($10\text{MW} \times 4.8 \times 365 \times 0.85$)
- **Annual revenue:** ~7,625,800,000 RWF (~\$6.6 million USD)
- **Simple payback:** ~5.7 years (based on reported \$37.6M CAPEX)
- **25-year savings:** Approximately 114-171 billion RWF (~\$99-\$149 million USD)

6.4 Comparison with Conventional Power Generation

6.4.1 Diesel Generation Comparison

- **Diesel generator efficiency:** ~3-4 kWh/L
- **Fuel cost:** 1,700 RWF/L
- **Effective cost:** ~425-567 RWF/kWh
- **Solar advantage:** Solar at 182 RWF/kWh provides 65-75% cost savings vs diesel

6.4.2 Grid Electricity Comparison

- **Current grid mix:** ~47% hydro, remainder thermal (diesel/heavy fuel oil)
- **Grid electricity cost:** Similar to tariff rate (182 RWF/kWh) for commercial/industrial
- **Solar advantage:** Locks in electricity cost for 25 years vs potential tariff increases

6.5 Sensitivity Analysis

6.5.1 Key Variables Affecting ROI:

1. **Installation cost:** $\pm 20\%$ changes payback by ± 0.6 years
2. **Solar yield:** ± 0.5 h/day changes payback by 0.3 years
3. **Electricity tariff:** $\pm 20\%$ changes payback by 0.6 years
4. **O&M costs:** $\pm 50\%$ changes payback by ± 0.2 years

6.6 Recommendations for Kigali City Implementation

6.6.1 Phased Approach:

1. **Pilot Phase (100kW-500kW):**
 - Target: Government buildings, schools, hospitals
 - Benefit: Demonstrates technology, builds local expertise
 - Expected payback: 2.6-3.8 years
2. **Commercial Phase (1MW-5MW):**
 - Target: Industrial zones, commercial complexes
 - Benefit: Significant cost savings for large consumers
 - Expected payback: 2.6-3.8 years
3. **Utility Phase (5MW-10MW+):**
 - Target: Grid-connected utility scale
 - Benefit: Peak shaving, renewable energy targets
 - Expected payback: 4-6 years (including grid interconnection)

6.6.2 Financing Considerations:

- **Available incentives:** Potential for green climate funds, World Bank support
- **PPA models:** Power Purchase Agreements can eliminate upfront CAPEX
- **Green bonds:** Increasing availability for renewable energy projects
- **Development finance:** AfDB, IFC, FMO offer favorable terms for African solar

6.7 Conclusion

Solar PV power generation in the 100kW to 10MW range presents attractive ROI for Kigali City applications, with simple payback periods ranging from 2.6 to 5.7 years depending on system size and specific project costs. The excellent solar resource (4.8 kWh/m²/day), high electricity tariffs, and declining solar costs make solar PV increasingly competitive with conventional generation.

For immediate implementation, 100kW-500kW systems targeting high-tariff commercial and institutional customers offer the fastest returns. Larger utility-scale projects (5MW+) require more sophisticated financing but provide substantial long-term savings and contribute significantly to Rwanda's renewable energy goals.

6.8 Sources

1. Rwanda Solar Panel ROI Calculator | AfroTools
2. Gesto Energy | Rwanda 10MW Solar PV Development
3. Utility-Scale Solar Cost Per MW 2026: EPC Bid Guide
4. Solar - Rwanda Energy Group
5. AFROTools Estimate solar payback in Rwanda

Note: All currency conversions based on approximate 2026 exchange rates. Actual project costs may vary based on specific site conditions, financing structures, and market fluctuations.

7 Geothermal Energy in Rwanda: Summary of Findings

7.1 Overview

Rwanda has significant geothermal energy potential as part of its strategy to diversify energy sources, reduce fossil fuel dependence, and achieve universal electricity access.

7.2 Key Potential Estimates

- **Conservative estimate:** Up to 100 MW (Rwanda Energy Group, African Geothermal Association)
- **Higher estimates:** Some sources suggest potential exceeding 700 MW (Rwanda Dispatch)

7.3 Prospect Zones

7.3.1 Northwestern Volcanic Area

- Gisenyi
- Karisimbi
- Kinigi

7.3.2 Southern Rift-Fault Area

- Bugarama

7.3.3 Additional Areas Mentioned

- Near Lake Kivu
- Volcanoes National Park
- Rubavu Graben
- Mashyuza

7.4 Exploration History

- **1980s:** Early geothermal work began
- **2006:** Two deep wells drilled at Karisimbi (no viable system found)
- **2006-2012:** Early surface studies identified four prospects
- **Current focus:** Detailed surface investigations in Bugarama and Gisenyi, with gradient-well work in Bugarama

7.5 Development Plans (Per 2015 Master Plan)

- **5-year exploration phase:** Surface work, drilling, well testing
- **3-year construction phase:** If resources proven
- Rwanda planned to host the Seventh Africa Rift Geothermal Conference (ARGeo-C7) in October 2018

7.6 Strategic Importance in National Energy Policy

7.6.1 Electricity Access Goals

- Target: 100% electricity access by 2024
- Breakdown: 52% on-grid, 48% off-grid

7.6.2 Capacity Expansion Targets

- Increase installed capacity from 216 MW to 512 MW by 2024
- Geothermal positioned as clean, reliable baseload power

7.6.3 Climate Commitments

- Critical for meeting Rwanda's 2030 emissions-reduction pledge of 38%
- Geothermal provides "stable and non-emitting power" compared to other renewables

7.7 Investment Requirements

- **\$600 million:** Estimated needed for universal electricity access
- **\$11 billion:** Over next decade for mitigation and adaptation
- Private sector and financial institutions viewed as essential partners for development

7.8 Regional Cooperation and Lessons from Kenya

Rwanda can significantly accelerate its geothermal development by leveraging Kenya's extensive experience, as Kenya has emerged as a global leader with over 900 MW of installed geothermal capacity (primarily from the Olkaria complex) as of 2026.

7.9 Key Lessons from Kenya's Geothermal Success:

1. **Phased Exploration Approach:** Kenya uses slim holes (6-8.5") for initial exploration, followed by production wells (9.5-12.25") after resource confirmation, reducing wasted expenditure.
2. **Directional Drilling:** Targeting specific fracture zones increases well productivity by 30-50% compared to vertical wells.

3. **Reservoir Management:** Sophisticated monitoring including pressure/temperature tracking, tracer testing, and production profiling optimizes steam field longevity.
4. **Power Plant Standards:** Modular designs (150-350 MW units) allow for phased expansion and easier maintenance.
5. **Financial Mechanisms:** Partial risk guarantees from development partners (World Bank, AfDB) cover exploration drilling risks, while inflation-indexed PPAs provide revenue certainty.

7.10 Recommended Kenya-Rwanda Cooperation Framework:

- **Technical Exchange Program:** Formal partnerships with KenGen and GDC for knowledge transfer
- **Training Initiatives:** Develop local Rwandan capacity through Kenyan training programs
- **Joint Ventures:** Explore partnership models for initial drilling and plant development
- **Regional Drilling Hub:** Leverage Kenyan drilling contractors to establish East African expertise
- **Financing Collaboration:** Joint applications to climate finance facilities (Green Climate Fund, etc.)

By implementing these lessons and fostering regional cooperation, Rwanda can reduce technical risks, improve project economics, and accelerate the timeline for commercial geothermal production.

7.11 Current Status (as of research completion)

- Exploration ongoing with focus on surface investigations
- No commercial geothermal production yet despite significant potential
- Test sites planned at Kinigi and Karisimbi for future development

7.12 Sources Consulted

1. Rwanda Energy Group - Geothermal Section
2. African Geothermal Association - Rwanda Profile
3. UNIDO Geothermal Fact Sheet for Rwanda (PDF)
4. Rwanda Dispatch - Geothermal Schemes Article

Note: This summary compiles information from multiple sources researched on September 6, 2026.

8 Energy Required to Pump Water to Kigali City

8.1 Overview

This document analyzes the energy requirements for pumping water to Kigali City, Rwanda, focusing on the elevation difference between water sources and the city, and providing calculations for the electrical energy needed per cubic meter of water delivered.

8.2 Key Elevation Data

- **Kigali City elevation:** 1,476 meters above sea level [Source: Elevation.maplogs.com]
- **Primary water source (Lake Mugesera/Karunge):** 1,300 meters above sea level [Source: Wikipedia - Lake Mugesera]
- **Static elevation head:** 1,476 m - 1,300 m = **176 meters**

8.3 Fundamental Physics of Water Pumping

The theoretical energy required to lift water is calculated using the formula:

$$\text{Energy (joules)} = \text{Volume (m}^3\text{)} \times \text{Density (kg/m}^3\text{)} \times \text{Gravity (m/s}^2\text{)} \times \text{Head (m)}$$

Where: - Density of water = 1,000 kg/m³ - Gravity = 9.81 m/s²

To convert joules to kilowatt-hours (kWh): **Energy (kWh) = Energy (joules) / 3,600,000**

Combining these, the theoretical energy per cubic meter per meter of head is: **Energy (kWh/m³/m) = (1,000 × 9.81) / 3,600,000 = 0.002725 kWh/m³/m**

8.4 Practical Energy Calculation (Including System Efficiency)

Real-world pumping systems have energy losses due to: - Pump inefficiency (typically 60-85% efficient) - Motor inefficiency (typically 85-95% efficient) - Pipe friction losses - Fitting and valve losses

The practical formula is: **Energy (kWh/m³) = (Head in meters) / (367 × Overall Efficiency)**

Where 367 is derived from: 3,600,000 / (1,000 × 9.81) = 367

Typical overall efficiency for water pumping systems (pump + motor + controls): **60-75%**

8.5 Energy Calculation for Kigali's Static Head

Using Kigali's 176-meter static head:

8.5.1 At 60% Overall Efficiency:

$$\text{Energy} = 176 / (367 \times 0.60) = 176 / 220.2 = \mathbf{0.80 \text{ kWh/m}^3}$$

8.5.2 At 70% Overall Efficiency:

$$\text{Energy} = 176 / (367 \times 0.70) = 176 / 256.9 = \mathbf{0.68 \text{ kWh/m}^3}$$

8.5.3 At 75% Overall Efficiency:

$$\text{Energy} = 176 / (367 \times 0.75) = 176 / 275.25 = \mathbf{0.64 \text{ kWh/m}^3}$$

8.6 Additional Energy Components: Friction Losses

The static head calculation above only accounts for lifting water against gravity. Actual systems require additional energy to overcome:

1. **Pipe friction losses:** Depends on pipe length, diameter, material, and flow rate
2. **Fitting losses:** Elbows, valves, tees, etc.
3. **Entry/exit losses:** At intake and discharge points

For municipal water supply systems, friction losses typically add **20-50%** to the static head requirement, depending on: - Distance from source to city - Pipe diameter and condition - Flow rate (higher flow = higher friction losses) - System age and maintenance

8.6.1 Estimated Total Head for Kigali System

Assuming moderate friction losses (30% additional head): - Static head: 176 meters - Friction losses: $176 \times 0.30 = 53$ meters - **Total dynamic head:** ~229 meters

Revised energy calculation (at 70% efficiency): $\text{Energy} = 229 / (367 \times 0.70) = 229 / 256.9 = \mathbf{0.89 \text{ kWh/m}^3}$

8.7 Context: Water Consumption and Energy Implications

8.7.1 Kigali's Water Demand

- Estimated daily water consumption: ~100,000 m³/day (varies by source and growth)
- Annual consumption: ~36.5 million m³/year

8.7.2 Annual Electricity Requirement for Pumping

At 0.89 kWh/m³: - Daily: $100,000 \text{ m}^3 \times 0.89 \text{ kWh/m}^3 = \mathbf{89,000 \text{ kWh/day}}$ - Annual: $36,500,000 \text{ m}^3 \times 0.89 \text{ kWh/m}^3 = \mathbf{32,485,000 \text{ kWh/year}}$ (~32.5 GWh/year)

8.7.3 Cost Implications

Using Rwanda's electricity tariff (~182 RWF/kWh or ~\$0.16 USD/kWh): - Daily pumping cost: 89,000 kWh $\times$ 182 RWF/kWh = **16,198,000 RWF/day** (~\$14,240 USD/day) - Annual pumping cost: 32,485,000 kWh $\times$ 182 RWF/kWh = **5,912,270,000 RWF/year** (~\$5.2 million USD/year)

8.8 Comparative Analysis

8.8.1 Comparison to Other Cities

- **Johannesburg, South Africa:** ~0.4-0.6 kWh/m³ (lower elevation gain)
- **Denver, Colorado, USA:** ~0.3-0.5 kWh/m³ (Rocky Mountain foothills)
- **Mexico City:** ~0.8-1.2 kWh/m³ (significant elevation gain ~1,000m)
- **La Paz, Bolivia:** ~1.2-1.8 kWh/m³ (very high elevation ~3,600m)

Kigali's requirement of ~0.68-0.89 kWh/m³ reflects its moderate elevation gain of ~176m from Lake Mugesera.

8.8.2 Comparison to Alternative Water Sources

If Kigali were to use sources at different elevations: - **From Nyabarongo River (~1,350m):** Head = 126m $\rightarrow$ ~0.49-0.64 kWh/m³ - **From deeper wells (~1,400m):** Head = 76m $\rightarrow$ ~0.30-0.39 kWh/m³ - **From higher sources (~1,500m):** Would require less pumping or could use gravity flow

8.9 Optimization Opportunities

8.9.1 1. System Efficiency Improvements

- **High-efficiency pumps:** Modern pumps can reach 85-90% efficiency
- **Premium efficiency motors:** IE3/IE4 motors 92-96% efficient
- **Variable frequency drives (VFDs):** Match pump speed to demand, saving 20-50% energy
- **Regular maintenance:** Clean impellers, aligned shafts, proper lubrication

8.9.2 2. Infrastructure Optimization

- **Pipe sizing:** Larger diameter reduces friction losses (higher CAPEX, lower OPEX)
- **Loop systems:** Reduce peak flows and friction losses
- **Leak detection and repair:** Reduce total volume needing pumping
- **Pressure management:** Reduce excess pressure in distribution network

8.9.3 3. Renewable Energy Integration

- **Solar PV pumping:** Excellent match for daytime water demand
- **Hydropower recovery:** Energy recovery from pressure reduction valves
- **Grid-tied systems with net metering:** Offset pumping costs with renewable generation

8.10 Comparison with Solar PV Generation Potential

From the SOLAR_ROI_ANALYSIS in this repository: - **100kW solar system** in Kigali generates ~148,920 kWh/year - This could pump approximately: $148,920 \text{ kWh/year} \div 0.89 \text{ kWh/m}^3 = \mathbf{167,000 \text{ m}^3/\text{year}}$ - Or about **457 m³/day** - roughly 0.46% of Kigali's estimated daily demand

8.10.1 Solar-Powered Pumping Scenarios

- **1MW solar system:** Could pump $\sim 4,570 \text{ m}^3/\text{day}$ ($\sim 4.6\%$ of demand)
- **5MW solar system:** Could pump $\sim 22,850 \text{ m}^3/\text{day}$ ($\sim 23\%$ of demand)
- **10MW solar system:** Could pump $\sim 45,700 \text{ m}^3/\text{day}$ ($\sim 46\%$ of demand)

This demonstrates that solar PV could offset a significant portion of Kigali's water pumping energy requirements, particularly during daylight hours when both solar production and water demand peak.

8.11 Recommendations for Kigali Water Supply

1. **Conduct detailed hydraulic modeling** to determine exact friction losses
2. **Implement energy audits** of existing pumping stations
3. **Prioritize high-efficiency equipment** in upgrades and replacements
4. **Consider solar PV integration** for daytime pumping operations
5. **Implement VFDs** on all major pumping stations
6. **Explore gravity-fed options** from higher elevation sources where feasible
7. **Monitor and benchmark** energy intensity (kWh/m^3) regularly for continuous improvement

8.12 Sources

1. Elevation of Kigali, Rwanda - MAPLOGS: https://elevation.maplogs.com/poi/kigali_rwanda.37590.html
2. Lake Mugesera elevation - Wikipedia: https://en.wikipedia.org/wiki/Lake_Mugesera
3. Karege Drinking Water Supply System - Wikipedia: https://en.wikipedia.org/wiki/Karege_Drinking_Water_Supply_System
4. Engineering ToolBox - Water Pumping Costs: https://www.engineeringtoolbox.com/water-pumping-costs-d_1527.html
5. Standard pumping energy calculation formulas (hydraulic engineering principles)

Note: Actual energy consumption will vary based on specific system design, maintenance condition, flow rates, and local factors. The calculations above provide a reasonable engineering estimate for planning and analysis purposes.
